Advanced Combustion via Microgravity Experiments (ACME)

BRE Radiometer Characterization Report

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PREFACE

Under the Space Flight Systems Development and Operations Contract (SpaceDOC), the National Aeronautics and Space Administration (NASA) has developed a modular, multi-user experimentation facility for conducting fluid physics and combustion science experiments in the microgravity environment of the International Space Station (ISS). This facility, called the Fluids and Combustion Facility (FCF), consists of two test platforms: the Fluids Integrated Rack (FIR) and the Combustion Integrated Rack (CIR). Also included in SpaceDOC are the required support efforts for Mission Integration and Operations, consisting of the Telescience Support Center (TSC) and Mission Integration and Planning (MIP).

The Advanced Combustion via Microgravity Experiments (ACME) is mini-facility designed to support Principal Investigators (PI) interested in studying various aspects of advanced combustion in a microgravity environment. ACME will be integrated into the Combustion Integrated Rack (CIR) onboard the International Space Station in the US Lab Module.

ACME will perform microgravity experiments to study fire suppression/flame extinguishment via 5 experiments: s-flame, Flame design, CLD Flame, E-Field Flames, and Burning Rate Emulator Flames as described in the ACME Baseline Description Document, ACME-DOC-004. ACME will allow investigation using pure as well as mixtures of gaseous fuels at various chamber atmosphere pressures, compositions, and oxygen concentrations. The microgravity environment allows for observation and measurement of flame formation and extinguishment processes that cannot be made in sustained gravity. The modular design of ACME allows for flexibility in configuration for specific experiments and easy maintainability of the hardware.

The core elements of gaseous combustion are included in the ACME design as a basis for current and future gaseous combustion studies. These elements are establishment of gas flow through various exchangeable mass flow controllers, ignition of gas at the burner and the use of appropriate diagnostic measurement devices. The five identified experiments, and any future experiments, use these core elements to build a unique set of requirements for the study of various phenomena distinctive to gaseous combustion. The goal of ACME is to improve the fundamental understanding of combustion processes in order to enable increased efficiency and reduce pollutant emission in practical combustion.

SIGNATURES / APPROVALS

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1.0 INTRODUCTION

1.1 Purpose

The purpose of this report is to document the results for the characterization of the BRE Radiometer Assembly (S1101MFA1550), Flight unit, S/N 1002.

1.2 Scope

The scope of this report covers all characterizations, calibrations and tests that were performed in April, 2016 on the ACME flight model. Results and subsequent analysis will be recorded in this final characterization report. This document does not cover all verification test results, but may be used as a reference for verifications.

1.3 Classification

The classification for the BRE Radiometer Assembly is Flight Model, including all the hardware listed in Table II.

2.0 DOCUMENTS

This section lists specifications, models, standards, guidelines, handbooks, and other special publications. These documents have been grouped into two categories: applicable documents and reference documents.

2.1 Order of Precedence for Documents

In the event of a conflict between this document and other documents referenced herein, the information of this document shall apply. In the event of a conflict between this document and the contract, the contractual requirements shall take precedence over this document. All documents used, applicable or referenced are to be the issues defined in the Configuration Management (CM) contract baseline. All document changes, issued after baseline establishment, shall be reviewed for impact on scope of work. If a change to an applicable document is determined to be effective and contractually approved for implementation, the revision status will be updated in the CM contract baseline. The contract revision status of all applicable documents is available by accessing the CM database. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.2 Applicable Documents

The following documents are applicable to the ACME project to the extent specified herein.

S1101MFA1550 Rev. B	BRE Radiometer Assembly
ACME-REQ-001 Rev. J	ACME Engineering Requirements Document
ACME-TST-022 Rev. A	Radiometer Board Test Procedure
P13001	SpaceDOC Procedure for the Control of Nonconforming Product
P05002	Configuration Management Procedure
P30020	SpaceDOC Safety and Mission Assurance Plan

2.3 Reference Documents

The documents/drawings in this section are provided only as reference material for background information and are not imposed as requirements.

ANSI-ESD S20.20.2007	ESD Association for the Development of an Electrostatic Discharge Program for Protection of Electronic Parts, Assemblies and Equipment (excluding Electrically Initiated Explosive Devices)
ACME-DOC-002 Rev. D	ACME Compliance Matrix
ACME-DOC-007 Rev. A	ACME Concept of Operations
ACME-SRD-001 Rev. -	Integrated Science Requirements Document

3.0 FACILITIES, EQUIPMENT, AND INSTRUMENTATION

3.1 Test Facilities

The flight BRE Radiometer Assembly Characterization took place in April, 2016 in the ZIN Technologies lab facility located at 6749 Engle Road.

3.2 Equipment and Instrumentation

The following table lists the name and type of the equipment and instrumentation required for the BRE Radiometer Package Characterization. The calibration status of all instrumentation was verified as appropriate before the characterization began.

TABLE I. Equipment and Instrumentation Information

Equipment	Manufacturer	Model Number	Cal Due Date
DC Regulated Power Supply, 28 VDC	Kenwood	Model: PD36-20AD S/N: 5010016 ZIN Tag: Z000162	N/A (via voltmeter)
Digital Voltmeter/Multi-meter	Fluke	Model: 179 S/N: 81730258 ZIN Tag: Z000371	11 May, 2016
Radiometer Assembly Test Fixture	ZIN Technologies	S1101EGA4583 S/N: 1001	N/A
Laptop w/ Ethernet & Serial Port	Dell	Model: Latitude E5510 S/N: CR8W4N1 ZIN Tag: Z004622	N/A
Oscilloscope	Tektronix	Model: HPLS-36APL750 S/N: 0609520 NASA Tag: 43895	N/A
Blackbody Infrared Source	Infrared Systems Development	Model: IR-564 S/N: 3062K NASA Tag: 09083	7/2016
Blackbody Temp. Controller	Infrared Systems Development	Model: MSO-3014 S/N: C025276 ZIN Tag: 58723	7/2016

4.0 DESCRIPTION OF TEST ARTICLE & CONFIGURATION

4.1 Test Article

4.1.1 BRE Radiometer Assembly

The BRE Radiometer Assembly (S1101MFA1550) consists of three (3) independent commercial Thermopile detectors (Dexter Research Center, XP-0249 Wide Angle 2M) mounted on a single PC board assembly (BRE Radiometer Board Assembly, S1101EFA1553). This assembly is mounted in a single housing (BRE Radiometer Housing Assembly, S1101MFA1551), along with the associated harness for power and data (BRE Radiometer Harness Assembly, S1101EFA1554).

The BRE Radiometer Assembly flight unit was assembled in-house by ZIN Technologies at 6749 Engle Road.

TABLE II. Test Article

Description	Manufacturer	Drawing Number	S/N
BRE Radiometer Assembly	N/A	S1101MFA1550	1002

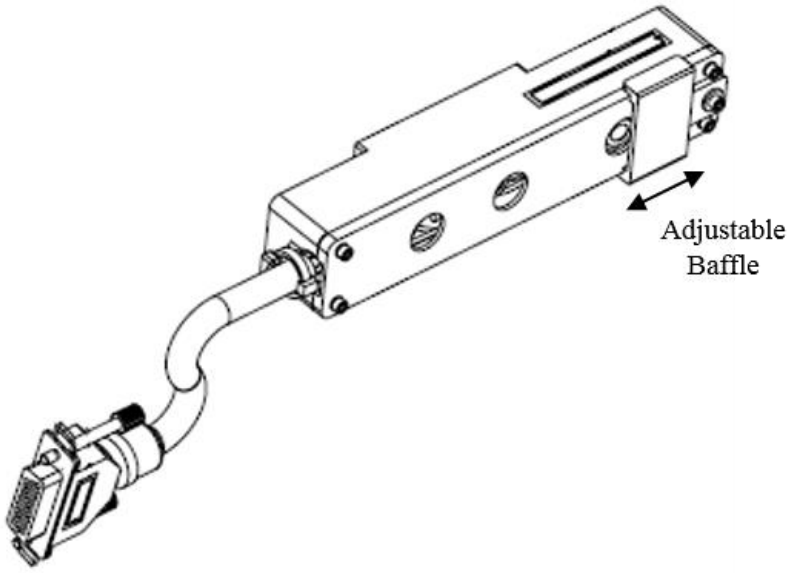


FIGURE 1. BRE Radiometer Assembly

The 3 detectors in the BRE Radiometer Assembly are mounted at distances of +60 mm, +30 mm and -10 mm from the centerline. In addition, the detectors are hard mounted at various angles and distances from the burn region, in order provide an approximate view of the same volume: a cylinder above the burner that is 50 mm diameter x 50 mm height.

The detector located at -10 mm is equipped with an adjustable baffle (see previous Figure) that slides up and locks into its location, in order to block the burner itself from the resulting irradiance detection.

The BRE Radiometer layout is shown in the Figure below.

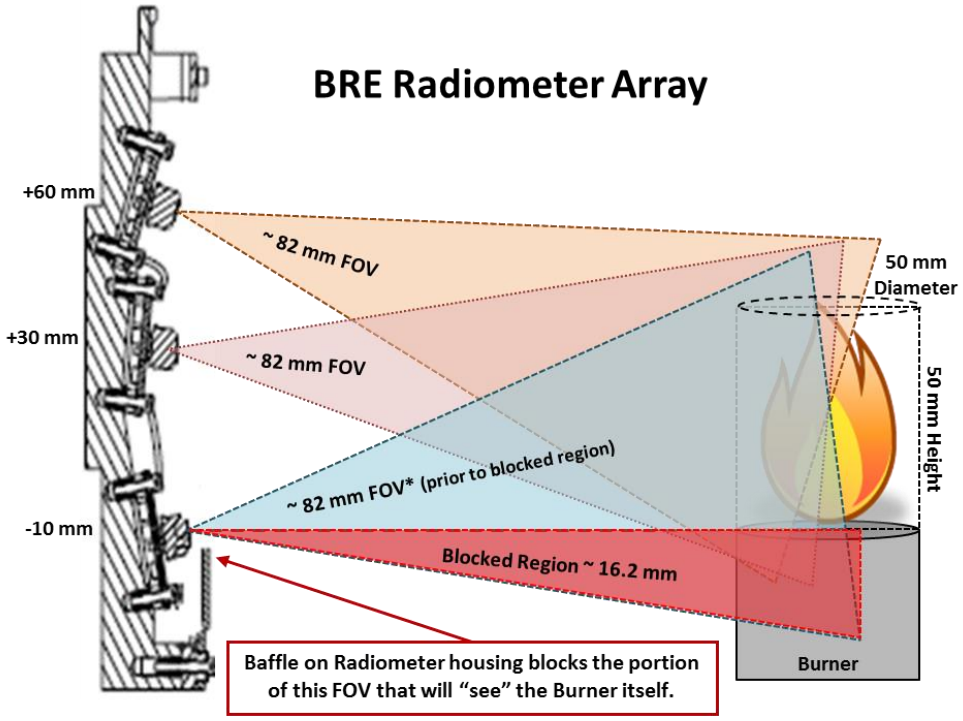


FIGURE 2. BRE Radiometer Layout

4.1.1.1 Thermopile Detectors – BRE Radiometer Assembly

Each of the 3 detectors in the ACME Radiometer Assembly is a commercial product made by Dexter Research Center. The Model 2M is a thin film based thermopile detector housed in a TO-5 metal package. The detector element provides a flat spectral response from 100 nm to 100 µm, with an active area of 4.0 mm^2. The back-fill gas is Argon, and the aperture size for the Wide Angle detectors is 0.144". All other Model 2M performance specifications supplied by Dexter Research Center are listed in the table below.

TABLE III. Dexter Research Center Model 2M Specs

Parameter	Min	Typical	Max	Symbol	Units
Active Area size		2 x 2		AA	mm
Element Area		4		A	mm^2
Number of Junctions		48			
Number of Channels		1			
Output Voltage	200	250	300	V_s	µV
Signal-to-Noise Ratio	12,739	19,531	30,303	SNR	√Hz
Responsivity	15.2	18.9	22.7	ℜ	V/W
Resistance	5	10	15	R	kΩ
Temperature Coefficient of ℜ		-.36			%/°C
Temperature Coefficient of R		-.2			%/°C
Noise Voltage	9.9	12.8	15.7	V_n	nV/√Hz
Noise Equivalent Power	.44	.68	1.0	NEP	nW/√Hz
Detectivity	1.9	3.0	4.6	D*	10^8cm√Hz/W
Time Constant		85		τ	ms
Field of View		38°/95°		FOV	Degrees
Package Type		TO-5			
Operating Temperature	-50		100	T_a	°C

4.1.1.2 Thermopile Window Material

The Window material selected for the detectors used is Barium Fluoride (BaF_2). This material selection is common to all of the thermopiles in the BRE Radiometer Assembly. Barium Fluoride has a typical average Transmission of 91% over a wavelength range of 0.2 – 17.5 µm. The transmission curve is shown in the figure below.

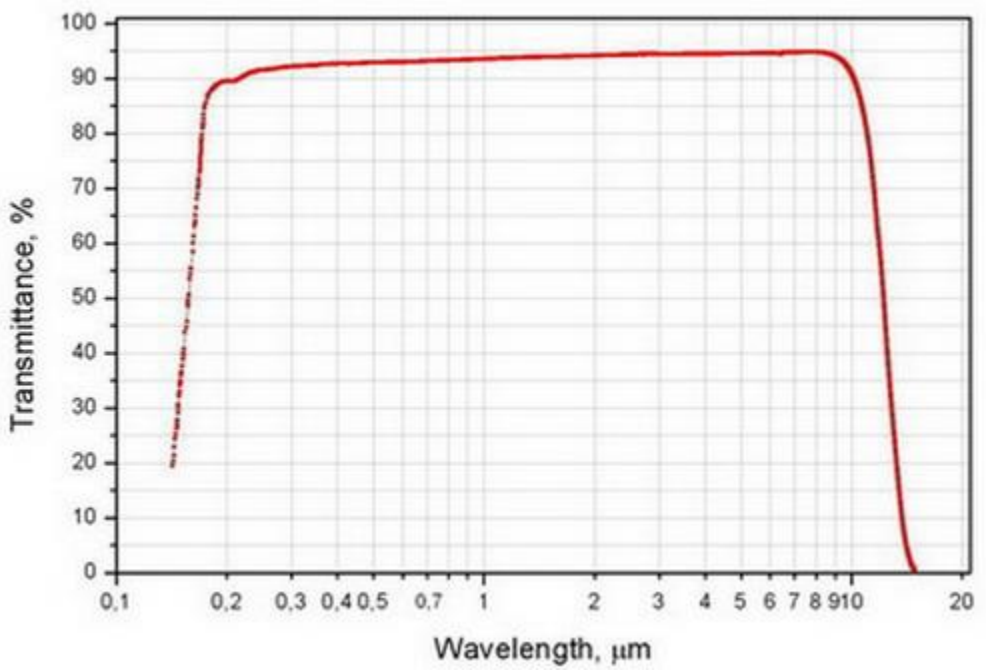


FIGURE 3. Barium Fluoride Transmission Curve

4.1.1.3 Test Configuration

BRE Radiometers are used for temperature based measurements, thus a calibrated Blackbody temperature source was used for this characterization.

The Radiometer Assemblies being characterized and the Blackbody source are positioned on an optical bench. The Blackbody source output was aligned, via optical mounting hardware, with the opening of the particular Radiometer detector location being characterized.

The optical mounting hardware allows for alignment of each Radiometer Assembly in both the X and Y directions and also sets the distance from the outer surface of each detector to the Blackbody aperture wheel (Z direction), equal to the appropriate flight distances of: 153.73 mm, 147.76 mm and 146.45 mm for the 3 detectors in the BRE Radiometer Assembly.

All testing was done under blackout curtains that were arranged such that only the output from the Blackbody was visible during characterization. See the picture below for the full setup (Note: side curtains are raised in order to obtain the picture).

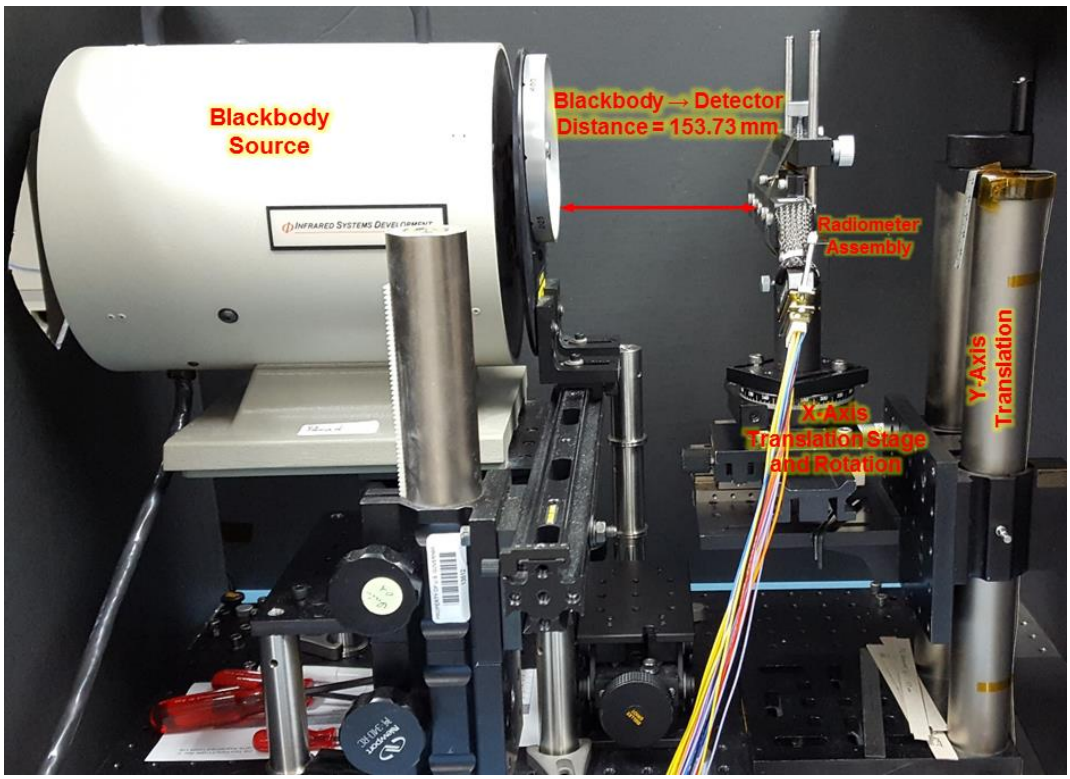


FIGURE 4. Radiometer Assembly Alignment Setup

To reduce the error caused by the Blackbody Aperture Wheel heating up over time and emitting unintended radiation, all Blackbody characterizations in this document were run with the Aperture Wheel mounted it on a slide rail, allowing it to be retracted away from the hot Blackbody source between measurements. By limiting the time resident near the hot Blackbody source, the aperture wheel could be held at room temperature.

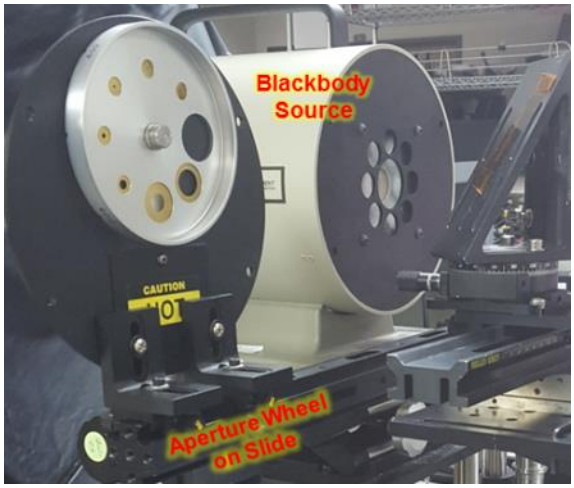


FIGURE 5. Aperture Wheel on Translation Slide

The BRE Radiometer Assembly is connected to the Radiometer Assembly Test Fixture for both +5 VDC power in and output (V_{out}) readings as shown in the figure below. Further details on the specific connections can be found in ACME-TST-022, "Radiometer Board Test Procedure."

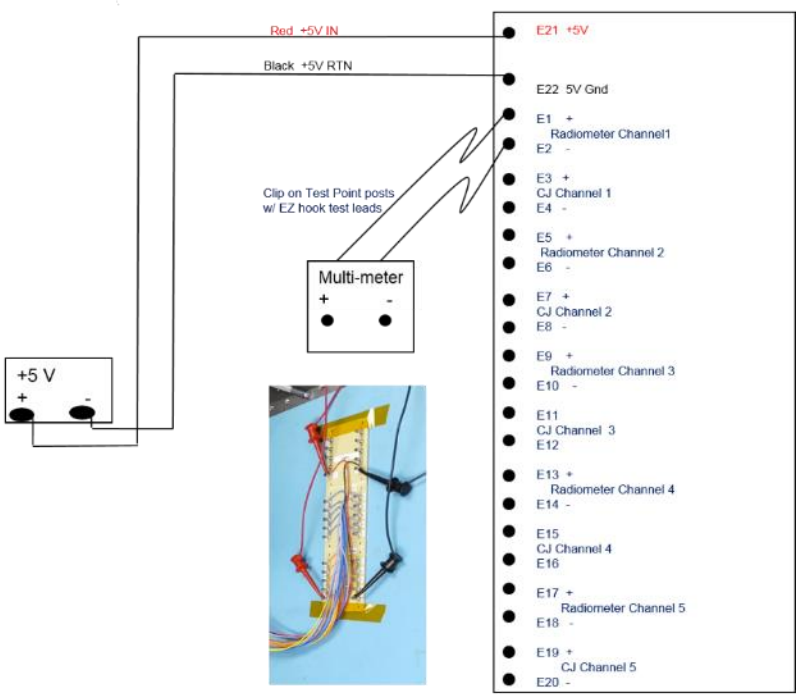


FIGURE 6. Test Fixture for Characterization Setup

A voltmeter was used to monitor the output signal (V_{out}) of each individual detector in the assembly. An additional voltmeter was connected to the Power Supply to ensure the input voltages remained within the required +5 VDC range. The DC Power Supply, Voltmeters, Oscilloscope and Blackbody Temperature Controller are located on a bench adjacent to the Radiometer Assembly and Blackbody (on the optical bench behind this shelf). Below is a photo of the support equipment in the laboratory setting.



FIGURE 7. GSE for Characterization Setup

5.0 BRE RADIOMETER ASSEMBLY CHARACTERIZATION

5.1 BRE Radiometer Assembly Field of View

5.1.1 FOV Setup

The Field of View (FOV) measurement is performed separately for each of the 3 thermopile detectors in the BRE Radiometer Assembly. Each detector is centered and aligned normal to the Blackbody source, at the appropriate flight distance from the Blackbody Aperture to the outer face. Based on expected FOV, each detector is then translated in the horizontal (X) axis over a range of ±60 mm about the center position, in 10 mm increments, while recording the output signal level. This determines the X axis FOV and roll-off characteristics.

While positioned at the X axis center point for that detector, the vertical (Y) axis characterization is measured by raising/lowering the assembly on a lab jack over ±70 mm to measure the Y axis FOV and roll-off characteristics.

5.1.2 +60 mm Radiometer Field of View Data / Analysis

Prior to the measurements, the +60 mm detector was set to its Flight Distance = 153.73 mm and aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for the +60 mm Radiometer.

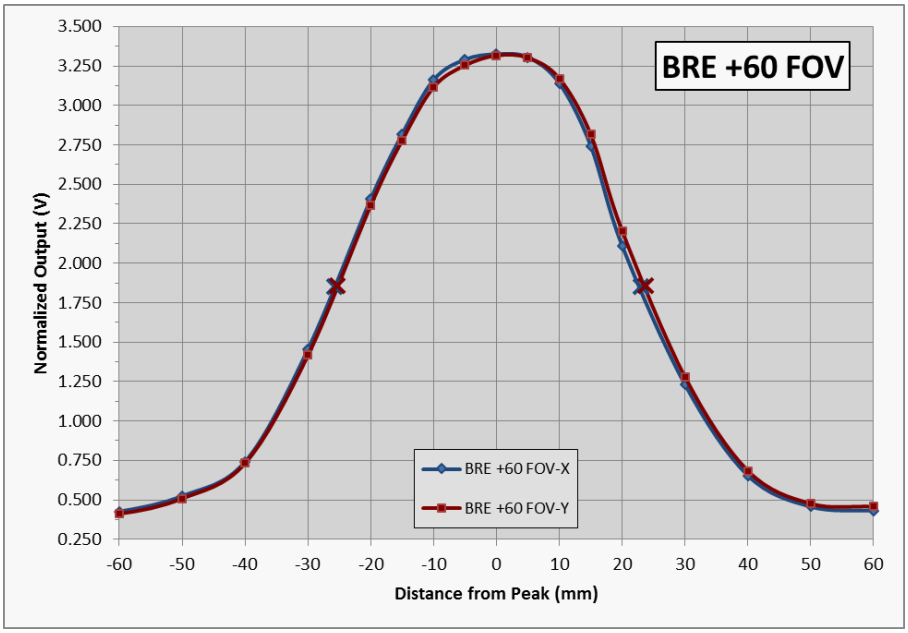


FIGURE 8. +60 mm – FOV Characterization

The following parameter settings were used for the +60 mm FOV scans:

Blackbody Temperature = 600°C
Blackbody Aperture = 0.4”

The following table shows the data resulting from the FOV scans for the +60 mm Radiometer.

TABLE IV. Data for +60 mm FOV

+60 - X Stats:			**+60 - Y Stats:**		
P-P	2.945	V	P-P	2.921	V
V_{FWHM}	1.853	V	V_{FWHM}	1.856	V
X1	-25.8	mm	Y1	-25.4	mm
X2	22.9	mm	Y2	23.8	mm
FWHM	48.8	mm	FWHM	49.1	mm
$Center_{FWHM}$	**-1.5**	**mm**	**$Center_{FWHM}$**	**-0.8**	**mm**
$V_{90\%}$	3.031	V	$V_{90\%}$	3.024	V
$X1_{90\%}$	-11.9	mm	$Y1_{90\%}$	-11.4	mm
$X2_{90\%}$	11.4	mm	$Y2_{90\%}$	12.1	mm
$FW_{90\%}$	23.3	mm	$FW_{90\%}$	23.4	mm
$Center_{Alignment}$	**-0.3**	**mm**	**$Center_{Alignment}$**	**0.3**	**mm**
V_{FOV}	0.675	V	V_{FOV}	0.687	V
$X1_{FOV}$	-43.1	mm	$Y1_{FOV}$	-42.0	mm
$X2_{FOV}$	39.6	mm	$Y2_{FOV}$	39.9	mm
FOV	**82.8**	**mm**	**FOV**	**82.0**	**mm**
$Center_{FOV}$	-1.7	mm	$Center_{FOV}$	-1.0	mm

The results for +60 mm Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of -1.5 mm and -0.8 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

+60 mm Alignment: X = - 0.3 mm and Y = 0.3 mm

The 10% points were used to determine the overall Field of View:

+60 mm Radiometer FOV: X = 82.8 mm and Y = 82.0 mm

5.1.3 +30 mm Radiometer Field of View Data / Analysis

Prior to the measurements, the +30 mm detector was set to its Flight Distance = 147.76 mm and aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for the +30 mm Radiometer.

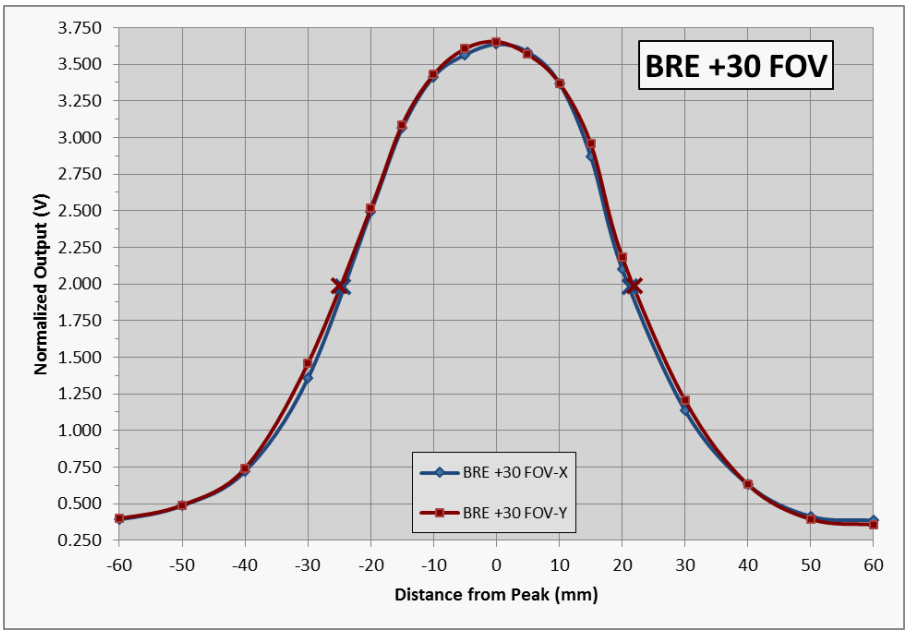


FIGURE 9. +30 mm – FOV Characterization

The following parameter settings were used for the +30 mm FOV scans:

Blackbody Temperature = 600°C
Blackbody Aperture = 0.4”

The following table shows the data resulting from the FOV scans for the +30 mm Radiometer.

TABLE V. Data for +30 mm FOV

+30 - X Stats:			+30 - Y Stats:		
P-P	3.304	V	P-P	3.331	V
V_{FWHM}	1.984	V	V_{FWHM}	1.989	V
X1	-24.5	mm	Y1	-25.0	mm
X2	21.2	mm	Y2	22.0	mm
FWHM	45.7	mm	FWHM	47.0	mm
$Center_{FWHM}$	**-1.6**	**mm**	**$Center_{FWHM}$**	**-1.5**	**mm**
$V_{90\%}$	3.306	V	$V_{90\%}$	3.321	V
$X1_{90\%}$	-11.6	mm	$Y1_{90\%}$	-11.6	mm
$X2_{90\%}$	10.7	mm	$Y2_{90\%}$	10.6	mm
$FW_{90\%}$	22.2	mm	$FW_{90\%}$	22.2	mm
$Center_{Alignment}$	**-0.5**	**mm**	**$Center_{Alignment}$**	**-0.5**	**mm**
V_{FOV}	0.662	V	V_{FOV}	0.656	V
$X1_{FOV}$	-42.5	mm	$Y1_{FOV}$	-43.4	mm
$X2_{FOV}$	39.4	mm	$Y2_{FOV}$	39.6	mm
FOV	**81.9**	**mm**	**FOV**	**83.0**	**mm**
$Center_{FOV}$	-1.6	mm	$Center_{FOV}$	-1.9	mm

The results for the +30 mm Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by both the X_{center} and Y_{center} fits of -1.6 mm and -1.5 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

+30 mm Alignment: X = -0.5 mm and Y = -0.5 mm

The 10% points were used to determine the overall Field of View:

+30 mm Radiometer FOV: X = 81.9 mm and Y = 83.0 mm

5.1.4 –10 mm Radiometer Field of View Data / Analysis

Prior to the measurements, the –10 mm detector was set to its Flight Distance = 146.45 mm and aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The X scan was a baseline scan without the baffle in place. It will be introduced later followed by an updated X scan. The following curves show the results from the FOV scans in both the X (Baseline) and Y directions for the –10 mm Radiometer.

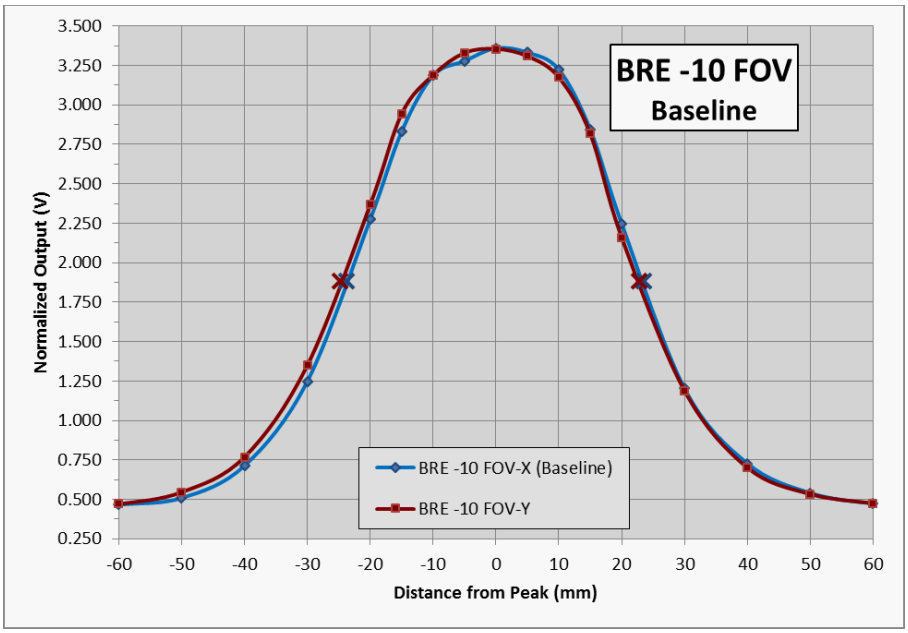


FIGURE 10. –10 mm – Baseline FOV Characterization

The following parameter settings were used for the –10 mm FOV scans:

Blackbody Temperature = 600°C
Blackbody Aperture = 0.4”

The following table shows the data resulting from the FOV scans for the –10 mm Radiometer, including the Y scan and the X Baseline scan with no baffle in place.

TABLE VI. Data for –10 mm Baseline FOV

-10mm - X (Baseline) Stats:			-10mm - Y Stats:		
P-P	2.955	V	P-P	2.938	V
V_{FWHM}	1.883	V	V_{FWHM}	1.886	V
X1	-23.8	mm	Y1	-24.7	mm
X2	23.5	mm	Y2	22.8	mm
FWHM	47.3	mm	FWHM	47.6	mm
$Center_{FWHM}$	**-0.2**	**mm**	**$Center_{FWHM}$**	**-1.0**	**mm**
$V_{90\%}$	3.065	V	$V_{90\%}$	3.061	V
$X1_{90\%}$	-11.7	mm	$Y1_{90\%}$	-12.6	mm
$X2_{90\%}$	12.1	mm	$Y2_{90\%}$	11.6	mm
$FW_{90\%}$	23.8	mm	$FW_{90\%}$	24.2	mm
$Center_{Alignment}$	**0.2**	**mm**	**$Center_{Alignment}$**	**-0.5**	**mm**
V_{FOV}	0.701	V	V_{FOV}	0.711	V
$X1_{FOV}$	-40.7	mm	$Y1_{FOV}$	-42.6	mm
$X2_{FOV}$	41.5	mm	$Y2_{FOV}$	39.8	mm
FOV	**82.1**	**mm**	**FOV**	**82.4**	**mm**
$Center_{FOV}$	0.4	mm	$Center_{FOV}$	-1.4	mm

The results for the –10 mm Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by both the X_{center} and Y_{center} fits of -0.2 mm and -1.0 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

-10 mm Alignment: X = 0.2 mm and Y = -0.5 mm

The 10% points were used to determine the overall Field of View:

-10 mm Radiometer FOV: X Baseline = 82.1 mm and Y = 82.4 mm

The –10 mm Radiometer Baseline X scan was used to establish the FWHM, alignment and overall FOV = 82.1 mm. From the calculations based on an 82 mm FOV, the region attributed to the burner that must be blocked was ~ 16 mm (see Figure 2).

Thus the X scan was repeated multiple times, adjusting the baffle location until the data produced a suitable profile showing that the calculated burner position would essentially be blocked from the FOV of the detector.

The following curves show the results from the FOV scans in both the X (Blocked) and Y directions for the –10 mm Radiometer with the baffle in its final location.

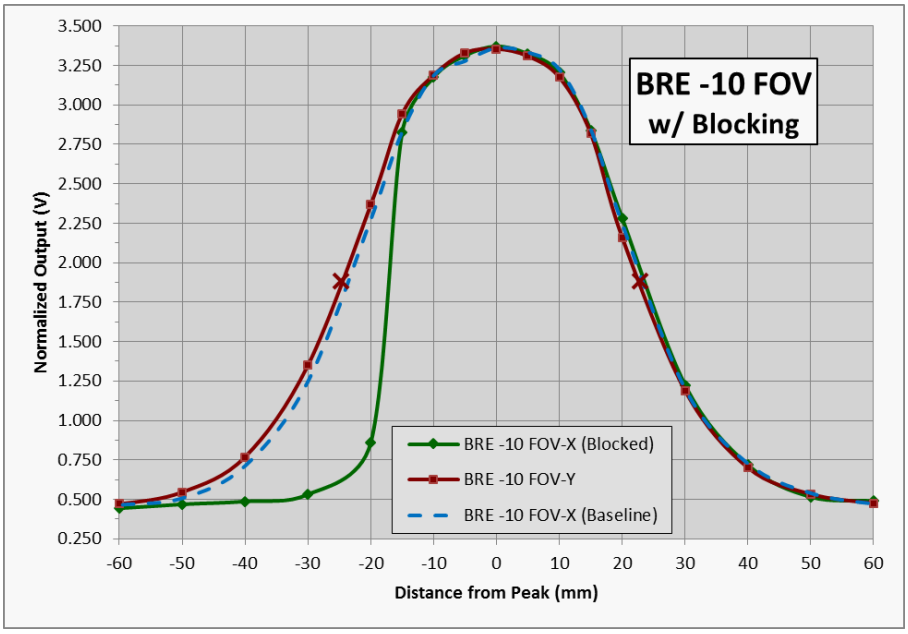


FIGURE 11. –10 mm – Blocked FOV Characterization

The following parameter settings were used for the –10 mm FOV scans:

Blackbody Temperature = 600°C
Blackbody Aperture = 0.4”

The following table shows the data resulting from the FOV scans for the –10 mm Radiometer, including the Y scan and the X Blocked scan with the baffle in its final location.

TABLE VII. Data for –10 mm Blocked FOV

-10mm - X (Blocked) Stats:		
P-P	2.952	V
V_{FWHM}	1.895	V
X1	-17.4	mm
X2	23.7	mm
FWHM	41.0	mm
$Center_{FWHM}$	**3.1**	**mm**
$V_{90\%}$	3.076	V
$X1_{90\%}$	-11.4	mm
$X2_{90\%}$	11.8	mm
$FW_{90\%}$	23.2	mm
$Center_{Alignment}$	**0.2**	**mm**
V_{FOV}	0.714	V
$X1_{FOV}$	**-24.4**	**mm**
$X2_{FOV}$	40.2	mm
FOV	**64.6**	**mm**
$Center_{FOV}$	7.9	mm
Blocked Region	**-16.2**	mm

-10mm - Y Stats:		
P-P	2.938	V
V_{FWHM}	1.886	V
Y1	-24.7	mm
Y2	22.8	mm
FWHM	47.6	mm
$Center_{FWHM}$	**-1.0**	**mm**
$V_{90\%}$	3.061	V
$Y1_{90\%}$	-12.6	mm
$Y2_{90\%}$	11.6	mm
$FW_{90\%}$	24.2	mm
$Center_{Alignment}$	**-0.5**	**mm**
V_{FOV}	0.711	V
$Y1_{FOV}$	-42.6	mm
$Y2_{FOV}$	39.8	mm
FOV	**82.4**	**mm**
$Center_{FOV}$	-1.4	mm

The results for the –10 mm Radiometer with the baffle of course show a skewed FOV, such that the center of symmetry looks to be +3.1 mm off in the X axis. Refer to the Baseline data for true Centering information.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

-10 mm Alignment: X = 0.2 mm and Y = -0.5 mm – Note these values still hold.

The 10% points were used to determine the overall Field of View, but again these values in the X axis were affected by to the blocking of a portion of the FOV:

-10 mm Radiometer FOV (w/ Blocking): X Blocked = 64.6 mm and Y = 82.4 mm

The Region that is blocked by the baffle was calculated using the final Baseline and Blocked FOV values:

Blocked X Region = -16.2mm mm

5.2 BRE Radiometer Assembly Radiometric Calibration

5.2.1 Radiometric Calibration Setup

The Radiometric Calibration is performed separately for each of the 3 detectors in the Radiometer Assembly.

Each detector in the assembly is first centered and aligned normal to the Blackbody source, at the appropriate Flight distance from the Blackbody Aperture to the outer detector face.

Then for each detector, the Blackbody source was stepped through temperatures ranging from 200°C – 1200°C (373°K - 1473°K). At each temperature, after the Blackbody was allowed to stabilize, the aperture wheel was moved into position in front of the Blackbody output and cycled through aperture sizes of 0.6", 0.4", 0.2", 0.1", 0.05" and 0.025" with Output Voltage (V_{out}) of the detector recorded at each. This simulates various source Irradiance levels covering the expected operating range. Readings were also taken with the aperture "closed" or covered in order to determine baseline readings for each Radiometer.

The Table below shows the expected Irradiance Range matrix for the calibration of the Radiometers.

TABLE VIII. Expected Irradiance Range for BRE Radiometer Assembly

Temp. (°C)	Aperture Diameter 0.6	0.4	0.2	0.1	0.05	0.025
200				Min 0.00002		
300			5% 0.00021			
400					Min 0.00002	
425	Max 0.0042		10% 0.00042			
500		50% 0.0021				
550				5% 0.00021		
600		Max 0.0042				
700				10% 0.00042		Min 0.00002
800			50% 0.0021			
900					5% 0.00021	
950			Max 0.0042			
1000						
1100					10% 0.00042	
1200				50% 0.0021		

5.2.2 +60 mm Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the +60 mm detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE IX. +60 mm Radiometric Calibration Data

+60	Aperture = 0.6"		Aperture = 0.4"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Aperture = 0.025"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	1.903	0.0006982	1.543	0.0003103	1.387	0.0000776	1.287	0.0000194					200	1.125
300	2.517	0.0015034	1.885	0.0006682	1.453	0.0001670	1.338	0.0000418					300	1.150
400	3.816	0.0028605	2.323	0.0012713	1.561	0.0003178	1.389	0.0000795	1.303	0.0000199			400	1.129
425	4.638	0.0038099	2.691	0.0016933	1.688	0.0004233	1.421	0.0001058	1.314	0.0000265			425	1.127
500	4.995	0.0053884	3.158	0.0022124	1.785	0.0005531	1.439	0.0001383	1.324	0.0000346			500	1.152
550	(Saturated)		3.778	0.0028427	1.932	0.0007107	1.467	0.0001777	1.347	0.0000444			550	1.111
600			4.486	0.0035989	2.061	0.0008997	1.503	0.0002249	1.363	0.0000562			600	0.927
700			4.995	0.0053884	2.488	0.0013883	1.612	0.0003471	1.405	0.0000868	1.309	0.0000217	700	1.012
800			(Saturated)		3.061	0.0020530	1.735	0.0005133	1.434	0.0001283	1.321	0.0000321	800	0.996
900					3.942	0.0029320	1.954	0.0007330	1.481	0.0001833	1.358	0.0000458	900	0.898
950					4.407	0.0034648	2.047	0.0008662	1.494	0.0002165	1.361	0.0000541	950	0.674
1000					4.995	0.0053884	2.193	0.0010168	1.512	0.0002542	1.372	0.0000635	1000	0.691
1100					(Saturated)		2.479	0.0013758	1.607	0.0003440	1.393	0.0000860	1100	0.738
1200							2.966	0.0018226	1.712	0.0004556	1.426	0.0001139	1200	0.746
													Avg =	0.963

The above data show the voltage range covers ~ 4.03 V, with the maximum output voltage, just below saturation of V_{max} = 4.99 V and the average minimum voltage, V_{min} = 0.963 V.

The measured Output Voltage range of 1.29 V – 4.99 V corresponds to an overall Irradiance Range of:

1.94×10^{-5} W/cm^2 to 4.21×10^{-3} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

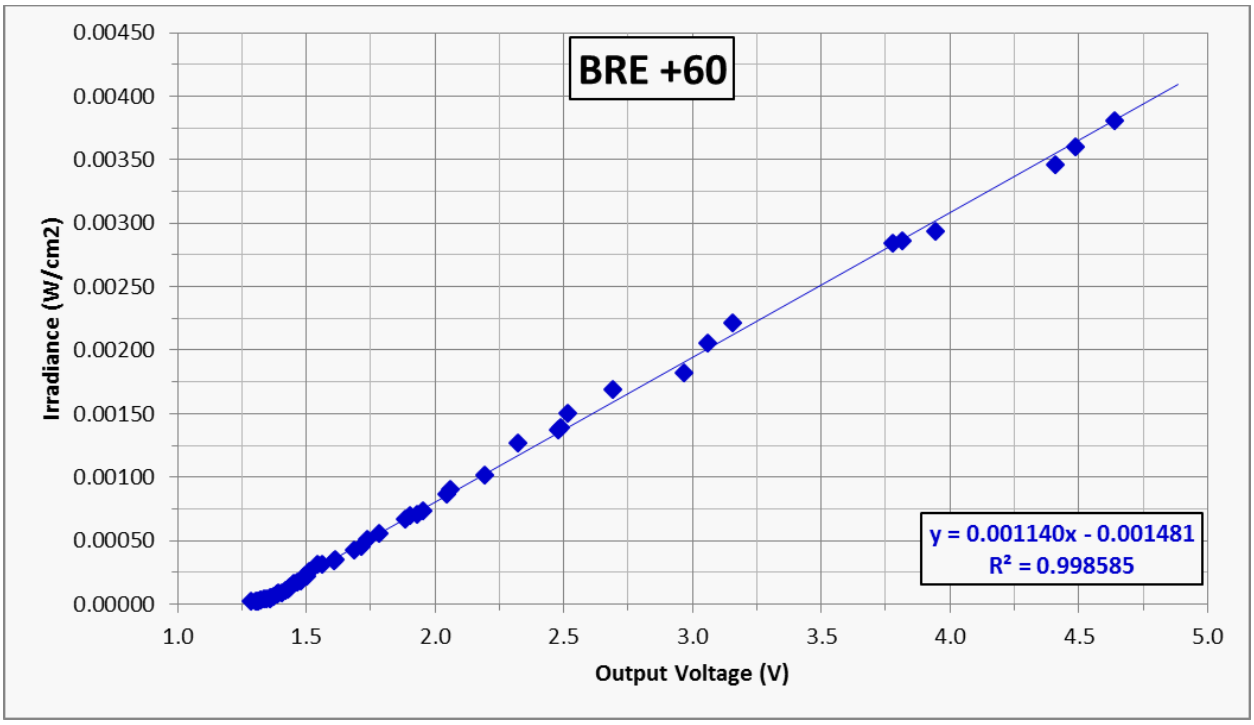


FIGURE 12. +60 mm Radiometric Data with Linear Fit

5.2.3 +30 mm Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the +30 mm detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE X. +30 mm Radiometric Calibration Data

+30	Aperture = 0.6"		Aperture = 0.4"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Aperture = 0.025"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	2.044	0.00075578	1.725	0.0003359	1.486	0.0000840	1.362	0.0000210					200	1.174
300	2.911	0.00162733	1.996	0.0007233	1.572	0.0001808	1.409	0.0000452					300	1.179
400	4.110	0.00309635	2.601	0.00137615	1.730	0.0003440	1.501	0.0000860	1.382	0.0000215			400	1.122
425	4.576	0.00358259	2.774	0.0015923	1.771	0.0003981	1.527	0.0000995	1.394	0.0000249			425	1.105
500	4.995	0.0053884	3.470	0.00239483	1.888	0.0005987	1.561	0.0001497	1.403	0.0000374			500	1.102
550	(Saturated)		4.078	0.0030771	2.071	0.0007693	1.593	0.0001923	1.429	0.0000481			550	1.096
600			4.821	0.0038956	2.248	0.0009739	1.654	0.0002435	1.474	0.0000609			600	1.093
700			4.995	0.0053884	2.717	0.0015027	1.753	0.00037568	1.524	0.0000939	1.393	0.0000235	700	1.079
800			(Saturated)		3.431	0.0022223	1.867	0.00055557	1.542	0.0001389	1.396	0.0000347	800	1.094
900					4.279	0.0031738	2.076	0.00079344	1.609	0.0001984	1.439	0.0000496	900	1.104
950					4.714	0.0037504	2.219	0.0009376	1.638	0.0002344	1.468	0.0000586	950	1.091
1000					4.995	0.0053884	2.387	0.00110057	1.674	0.0002751	1.483	0.0000688	1000	1.103
1100					(Saturated)		2.71	0.00148926	1.733	0.0003723	1.522	0.0000931	1100	1.116
1200							3.181	0.0019728	1.849	0.0004932	1.531	0.0001233	1200	1.109
													Avg =	1.112

The above data show the voltage range covers ~ 3.88 V, with the maximum output voltage, just below saturation of V_{max} = 4.99 V and the average minimum voltage, V_{min} = 1.112 V.

The measured Output Voltage range of 1.36 V – 4.99 V corresponds to an overall Irradiance Range of:

2.10×10^{-5} W/cm^2 to 4.04×10^{-3} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

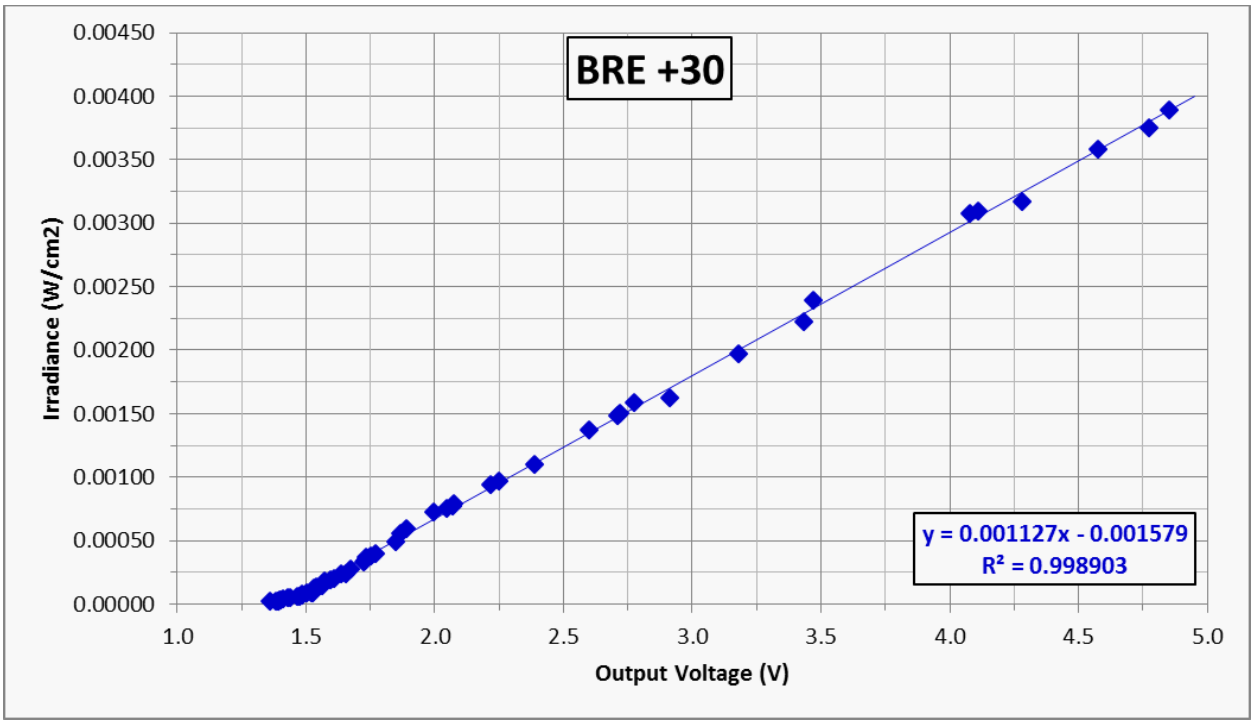


FIGURE 13. +30 mm Radiometric Data with Linear Fit

5.2.4 –10 mm Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the –10 mm detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XI. –10 mm Radiometric Calibration Data

-10	Aperture = 0.6"		Aperture = 0.4"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Aperture = 0.025"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	1.896	0.00075578	1.601	0.0003359	1.429	0.0000840	1.348	0.0000210					200	1.177
300	2.648	0.00162733	1.873	0.0007233	1.491	0.0001808	1.399	0.0000452					300	1.193
400	3.969	0.00309635	2.392	0.00137615	1.613	0.0003440	1.435	0.0000860	1.359	0.0000215			400	1.167
425	4.408	0.0035826	2.623	0.00159226	1.679	0.0003981	1.452	0.0000995	1.372	0.0000249			425	1.171
500	4.995	0.0053884	3.292	0.0023948	1.815	0.0005987	1.481	0.0001497	1.380	0.0000374			500	1.132
550	(Saturated)		3.898	0.0030771	1.931	0.0007693	1.512	0.0001923	1.406	0.0000481			550	1.125
600			4.624	0.0038956	2.113	0.0009739	1.542	0.00024348	1.416	0.0000609			600	1.107
700			4.995	0.0053884	2.556	0.0015027	1.638	0.00037568	1.445	0.0000939	1.363	0.0000235	700	1.023
800			(Saturated)		3.188	0.0022223	1.783	0.00055557	1.470	0.0001389	1.375	0.0000347	800	1.029
900					4.057	0.0031738	1.998	0.00079344	1.519	0.0001984	1.408	0.0000496	900	1.061
950					4.563	0.0037504	2.095	0.0009376	1.526	0.0002344	1.414	0.0000586	950	1.019
1000					4.995	0.0053884	2.227	0.00110057	1.583	0.0002751	1.419	0.0000688	1000	0.977
1100					(Saturated)		2.513	0.00148926	1.656	0.0003723	1.444	0.0000931	1100	0.761
1200							3.052	0.0019728	1.756	0.0004932	1.464	0.0001233	1200	0.812
													Avg =	1.054

The above data show the voltage range covers ~ 3.94 V, with the maximum output voltage, just below saturation of V_{max} = 4.99 V and the average minimum voltage, V_{min} = 1.054 V.

The measured Output Voltage range of 1.35 V – 4.99 V corresponds to an overall Irradiance Range of:

2.10×10^{-5} W/cm^2 to 4.30×10^{-3} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

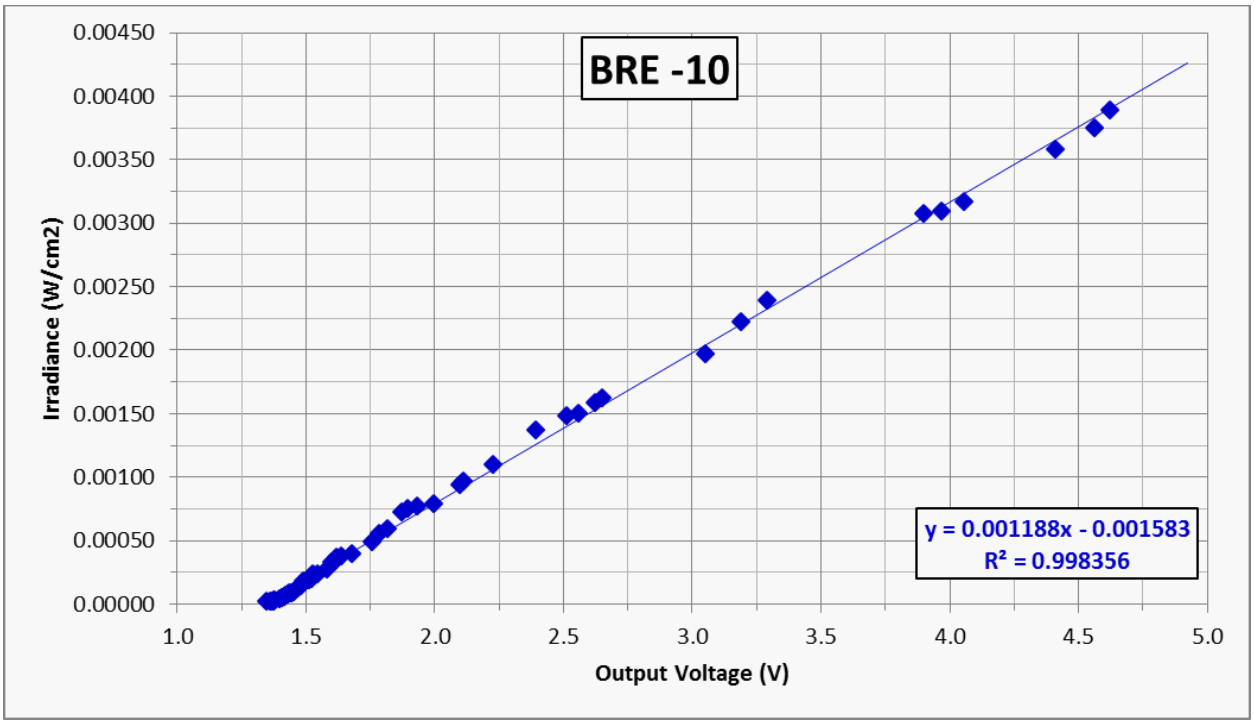


FIGURE 14. – 10 mm Radiometric Data with Linear Fit

5.3 Response Time

5.3.1 Response Time Analysis

The response time of each of the thermopile detectors in the BRE Radiometer Assembly was measured using an optical chopper wheel to provide a step function as irradiance input. The Output Voltage of each detector is then analyzed to calculate the time to rise from 0 – 62.5% of the signal range.

The optical chopper and controller used are ThorLabs Model MC1000. The wheel was configured for 2 cycles (i.e. 2 open regions and 2 blocked regions); hence a 1 Hz drive produces a 2 Hz response. The Figure below shows the results of the time response testing for all 3 of the detectors in the BRE Radiometer Assembly.

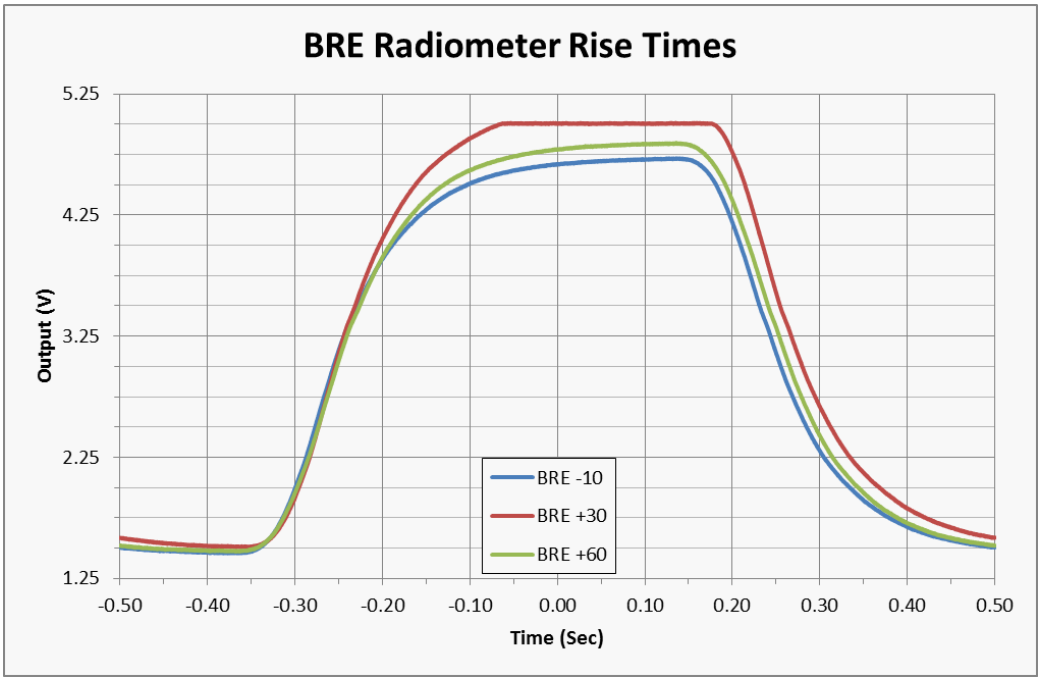


FIGURE 15. BRE Radiometer Assembly Time Response

The table below shows the calculated response time for each detector based on the method above.

TABLE XII. BRE Radiometer Assembly Response Time Values

	+60		+30		-10	
	V_{Out}	Time (s)	V_{Out}	Time (s)	V_{Out}	Time (s)
Min V	1.47	-0.03439	1.51	-0.23590	1.46	-0.35710
Max V	4.85		5.01		4.72	
Delta V	3.38		3.51		3.26	
62.50%	2.74	-0.25370	2.82	-0.14670	2.68	-0.26970
Time Delta		0.09200		0.08920		0.08740
Response Time		**90.2 ms**		**89.2 ms**		**87.4 ms**

5.4 Statistics & Error Analysis

The accuracy is determined by a combination of random and systematic errors accounted for in the accompanying statistical analysis. This analysis represents the graphical data by method of least squares. Note the calculated Irradiances in the plots are all subject to the uncertainties and the Output Voltages recorded in the calibration also have measurement uncertainties.

The graphical approach used does not inherently account for unknown systematic errors.

The Table below summarizes the results of the graphical analysis of the collected data for all 3 of the detectors in the BRE Radiometer Assembly.

TABLE XIII. BRE Radiometer Assembly Statistical Analysis

Radiometer Statistics	+60	+30	-10
Voltage Range (V)	3.35	3.49	3.28
Min Voltage (V)	1.29	1.36	1.35
Max Voltage (V)	4.64	4.85	4.62
Irradiance Range (W/cm^2)	0.0037905	0.0038746	0.0038746
Min Irradiance (W/cm^2)	0.0000194	0.0000210	0.0000210
Max Irradiance (W/cm^2)	0.0038099	0.0038956	0.0038956
Uncertainty (± V)	5.8954E-06	5.5952E-06	6.6206E-06
(Linear) R^2	**0.9986**	**0.9987**	**0.9984**
(Linear) Slope	0.001140	0.001126	0.001188
(Linear) Intercept	-0.001481	-0.001578	-0.001583

The Table below shows the final Total Error at the Minimum Irradiance and Maximum Irradiance values, as well as at values representing 5%, 10% and 50% of the Max value.

In addition, the Irradiance at the 5% Error threshold has been calculated and is displayed.

TABLE XIV. BRE Radiometer Assembly Accuracy & Error Analysis

Accuracy/Error Summary	+60	+30	-10
Error at Min	14.04%	13.32%	15.76%
Error at 5% Max	2.81%	2.66%	3.15%
Error at 10% Max	1.40%	1.33%	1.58%
Error at ½ Max	0.28%	0.27%	0.32%
Error at Max	0.14%	0.13%	0.16%
Min Irradiance @ 5% Error	0.0001179	0.0001119	0.0001324

APPENDIX A ACRONYMS AND ABBREVIATIONS

ACME	Advanced Combustion via Microgravity Experiments
BB	Broadband
CIR	Combustion Integrated Rack
COTS	Commercial off The Shelf
DC	Direct Current
ERD	Engineering Requirements Document
ESD	Electrostatic Discharge
FCF	Fluids and Combustion Facility
FWHM	Full Width at Half Max
FOV	Field of View
FS	Fused Silica
GL	Gray Level
IR	Infrared
ISS	International Space Station
ms	Milliseconds
N/A	Not Applicable
NASA	National Aeronautics and Space Administration
NB	Narrowband
ns	Nanoseconds
QE	Quantum Efficiency
SpaceDOC	Space Flight Systems Development and Operations Contract
UV	Ultraviolet
VDC	Volts (direct current)